

HASSLE-FREE COLLEGE ADMISSION PORTAL

*A Real-time Research Project report submitted in partial fulfilment of the
requirements for the award of the degree of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING(AI&ML)

by

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Department of Computer Science & Engineering (AI&ML)

VAAGESWARI COLLEGE OF ENGINEERING

Accredited by NAAC with A+ Grade

(Affiliated to JNTU Hyderabad & Approved by AICTE New Delhi)

Ramakrishna colony, Karimnagar-505527

JUNE 2025

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CERTIFICATE

This is to certify that the project report entitled “**HASSLE-FREE COLLEGE ADMISSION PORTAL**” submitted by the following students in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in CSE(AI&ML) and is a Bonafide record of the work performed by.

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DECLARATION

We hereby declare that the project titled “**HASSLE-FREE COLLEGE ADMISSION PORTAL**” submitted to Vaageswari College Of Engineering, affiliated to Jawaharlal Nehru Technological University Hyderabad (JNTUH) for the award of the Degree of Bachelor of Technology in CSE(AI&ML) is a result of original research carried-out in this work. It is further declared that the report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

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ABSTRACT

The project “university admission system” a web application. It is the best platform for the students who deserved to study higher education in their respective fields .The main objective of this project is to give the seat to the students in their desired University .This can be done through certain process. The two main components contain are user and administrator .user mainly gives the details to the respective questions asked by application. And other component administrator Gives the details to the application. As it was online based platform it is very comfort for the students to look over different universities and give their choices to the application to get placed in respective University.

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CHAPTER-1

INTRODUCTION

INTRODUCTION:

The Introduction to the University Admission System project sets the stage for a transformative initiative in modernizing and enhancing the traditional admission process. Recognizing the need for efficiency, transparency, and accessibility, this project endeavors to create a comprehensive online platform. By leveraging cutting-edge technology, the system aims to simplify application procedures, reduce manual intervention, and provide a seamless experience for prospective students and university administrators alike. This introduction outlines the rationale, objectives, and anticipated benefits of implementing a robust and user-friendly admission system, reflecting the commitment to advancing academic processes in today's digital age.

University Admission System marks a pivotal advancement in the realm of higher education administration. This innovative system emerges as a solution to the challenges posed by traditional admission processes, aiming to revolutionize the way universities admit students. By leveraging technology, this system not only enhances the efficiency of the application process but also fosters transparency and accessibility. In this era of digital transformation, the University Admission System stands as a beacon, offering a streamlined and user-friendly experience for both applicants and academic institutions.



CHAPTER-2

LITERATURE SURVEY

1.Title: online platform for university admission

Authors: Loan Filip, Andrea robu

Abstract: The admission process for numerous universities requires an entrance exam in order to select the most competitive students. The paper presents the design and implementation strategies, the architecture and the functionalities of a software application, called SimEntranceExam, developed for online simulation of admission exam. Besides its primary objective, the platform offers an adequate learning and testing environment, enhancing knowledge and self-evaluation process. It also presents the advantages of the online preparation for the admission exam, as opposed to the classical method of resolving paper based tests. The application contributes substantially to the improvement of the educational process.

2. Title: Research on college party' admission management system based on workflow

Authors: Shen zinao, Wang Hui

Abstract: This paper describes the workflow management system and its functional characteristics, gives the formal definition of workflow model. Combined with the present situation of college Party's admission, it discusses the business process of Party's admission, builds a Petri net-based workflow model. Based on Web applications as background, it presents solutions for college Party's admission management system, which satisfies the needs of college Party's admission. The schema with good usability and versatility has a certain application value.

3. Title: Accelerating university admission system using machine learning techniques

Authors: Hanadi Alazmi, Bansil altothman

Abstract: In this paper, we proposes a solution to enhance university admission system using different machine learning techniques like K-Nearest Neighbour(KNN), Decision Tree(DT) and Random Forest(RF). A deterministic algorithm is used to create a basis for the analysis, followed by several probabilistic algorithms, KNN, decision tree, and random forest. The outcomes are compared and a consensus is reached in categorising each student whether they got accepted, placed in foundation or rejected

in the desired institute. Processing university applications is time-consuming for admission staff members, it is not efficient in terms of cost and time, thus it can sometimes be an overwhelming task, the chances of bribes occurring, or discrimination is highly likely to occur. The solution for this problem is to automate the application process thus using the help of a software with several embedded AI algorithm that improves the efficiency and reliability of the application process.

4. Title: A quality based automated admission system for educational domain

Authors: Sushruta Mishra, Sowmya sahoo

Abstract: In last two decades several educational institutes have started gaining momentum while many of them are in self financing mode. Every institute wants to have good student strength to allow a smooth academic session. This paper proposes the use of machine learning techniques in educational domain to enhance the quality of student admissions in any higher educational institute. The focus of this paper is to identify those admissions inquires which most likely to turn into actual admissions. The result of analysis will assist the academic planners to focus their efforts on the set of students that are likely to take admission in the institution after initial enquiry.

5. Title: A university admission prediction system using stacked ensemble learning

Author: Sashank Sridhar, Siddhartha Motta

Abstract: For an aspiring graduate student, shortlisting the universities to apply to is a difficult problem. Since an application is extremely dynamic, students often tend to wonder if their profile matches the requirement of a certain university. Moreover, the cost of applying to a university is extremely high making it critical that students shortlist universities based on their profile. A university admission prediction system is quite useful for students to determine their chances of acceptance to a specific university. The system could make use of data related to previous applicants to various universities and their admit or reject status. Earlier models of such prediction systems suffer from several drawbacks such as not considering important parameters like GRE (Graduate Record Exam) scores or research experience. Further, the accuracy reported by earlier models is also not sufficiently high. In this paper, a stacked ensemble model that predicts the chances of admit of a student to a particular university has been proposed. The proposed model takes into consideration various factors related to the student

including their research experience, industry experience etc. Further, the system proposed has been evaluated against various other machine learning

CHAPTER-3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM:

The existing university admission system mainly gives the motive on how the existing system gives proper

Importance on University admission system. there are certain disadvantages how it exists in it. there are certain disadvantages.

DISADVANTAGES:

- **Paper-Based Processes:** Many systems still rely on traditional, paper-based methods, leading to inefficiencies, delays, and increased chances of errors.
- **Limited Accessibility:** Some systems lack user-friendly interfaces and may not be easily accessible to all applicants, potentially excluding those with limited internet access or technological literacy.
- **Manual Data Entry:** Manual data entry can result in data inaccuracies and consume considerable time and resources.

Inefficient Communication: Ineffective communication channels between the university and applicants can hinder the dissemination of important information.

3.2 PROPOSED SYSTEM:

A proposed university admission system aims to overcome the existing system disadvantages. It gives certain importance and certain new implementation on existing system. There are certain advantages according to point wise.

ADVANTAGES:

- **Online Application:** Implementing an online application system facilitates a seamless and user-friendly experience for applicants, enabling them to submit materials and track their progress easily.
- **Automation of Processes:** The system automates various admission processes, reducing manual workload, minimizing errors, and expediting the overall admission timeline.

- **Increased Accessibility:** An online platform enhances accessibility, allowing applicants to submit documents and check their application status from anywhere with an internet connection.
- **Data Accuracy:** Automation reduces the likelihood of errors in data entry, ensuring accurate information is processed throughout the admission cycle.
- **Transparency:** The system provides transparency in the admission process, offering real-time updates and clear communication to applicants about their status and any additional requirements.

CHAPTER-4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS:

- PROCESSOR: I3.
- Hard Disk : 256 GB.
- Ram : 8 GB.

4.2 SOFTWARE REQUIREMENTS:

- Operating system : Windows.
- Coding Language : JAVA/J2EE
- Data Base : MYSQL
- IDE :Netbeans8.1

CHAPTER-5

SYSTEM STUDY

5.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

5.2 FEASIBILITY ANALYSIS

Three key considerations involved in the feasibility analysis are

- ◆ ECONOMICAL FEASIBILITY
- ◆ TECHNICAL FEASIBILITY
- ◆ SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY:

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY:

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system

CHAPTER 6

SYSTEM DESIGN

SYSTEM ARCHITECTURE

The design and implementation of computerized students' admission system is focused at realizing an efficient computation of students' results. These complaints arise from problems discovered in the manual system which not only include inefficiency, inaccuracy, time consumption and inconvenience but laborious and sluggish. The proposed model is focused with studentship monitoring, storing students' fundamental academic records, tracking students if there is any form of wrongdoing, and checking every certificate issued by Ghana Communication Technology University in the future to ensure that false admittance is greatly decreased. An institution cannot function without information; hence an information system is similar to the nervous system in the human body. In this project, a review of other project works was performed and analyzed, giving the same problems listed in the project. Chapter three is a systematic design on how to execute a web design for the admission portal. ER Diagram and Design model were introduced to explain the details of the admission portal. In Chapter 4, there was an implementation of the designed program. The admission portal was administered in three stages, the first stage which is the first page is the login page. The second stage is the admission requirement and the documents needed to complete the admission form and the third stage is where the admission form is downloaded. After downloading the admission form, applicants are supposed to fill them up and upload them unto the same page, including the documents required for the admission form. After that, there is a submit button to submit all documents. This program was designed using PHP and HTML for the frontend design and DJANGO for the backend design. This program will eradicate the problems that both staff and students face.

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major

components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

6.2.1 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

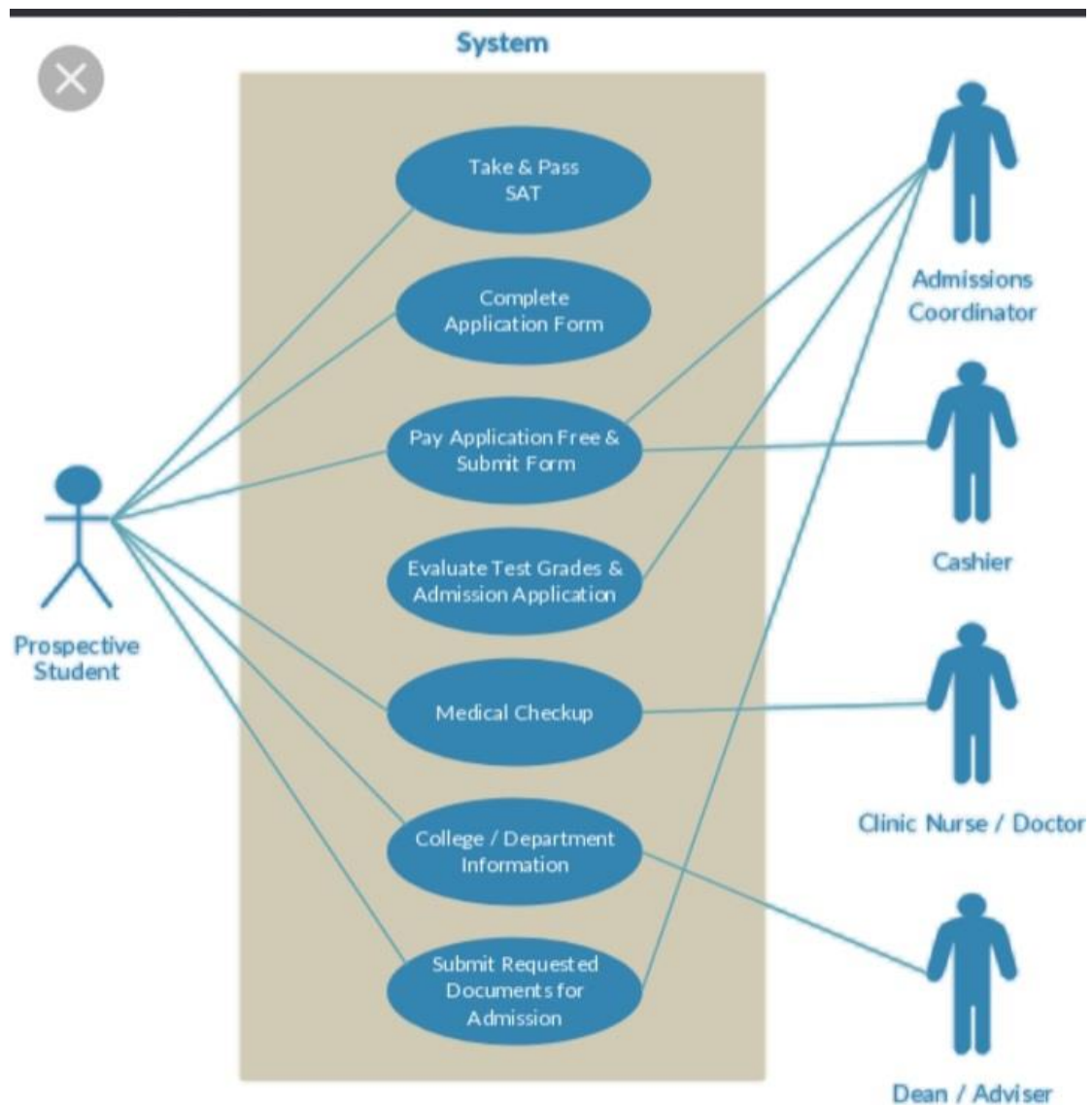


Fig 6.2.1.1: Use Case Diagram

6.2.2 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information. Class diagrams are a fundamental tool in object-oriented modeling and are widely used in software development.

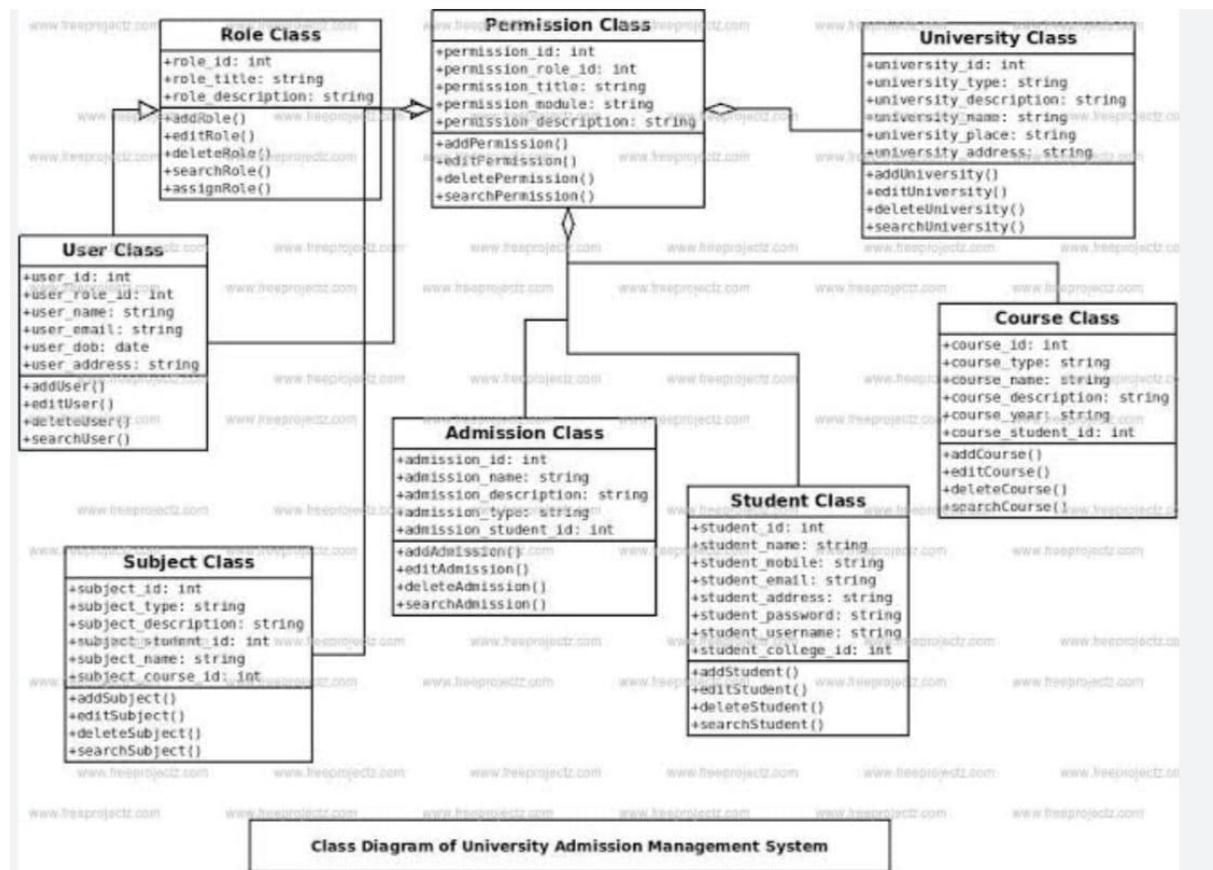


Fig 6.2.2.1: Class Diagram

6.2.3 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

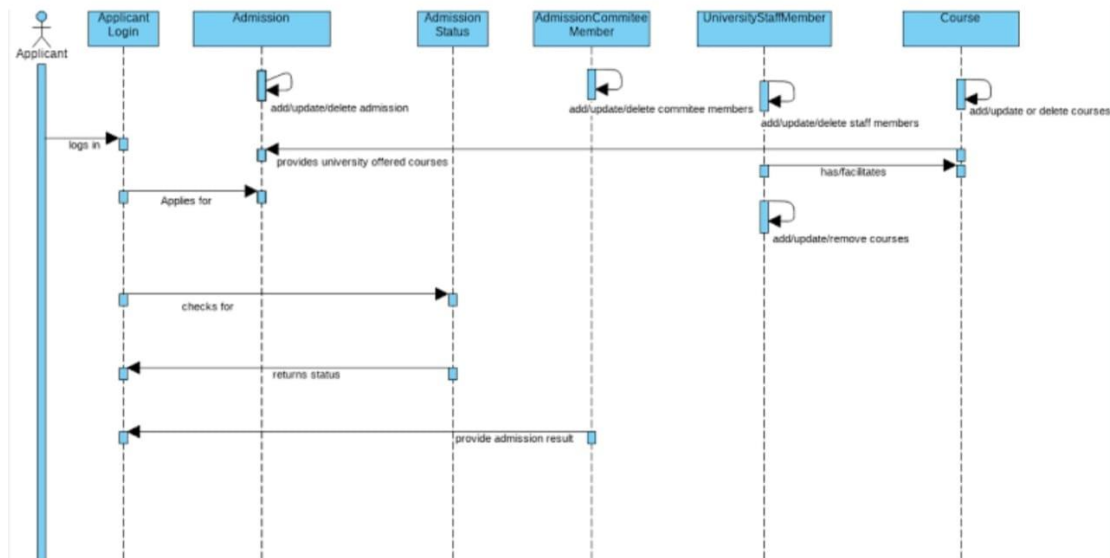


Fig 6.2.3.1: Sequence Diagram

6.2.4 COLLABARATION DIAGRAM:

A Collaboration diagram, also known as a Communication diagram, is a type of Unified Modelling Language (UML) diagram used to visually represent the interactions and communication between objects or components in a system or software application. It focuses on showing how various objects or components collaborate to achieve specific functionality or a particular use case. Collaboration diagrams are particularly useful for understanding the dynamic aspects of a system and how objects interact over time.

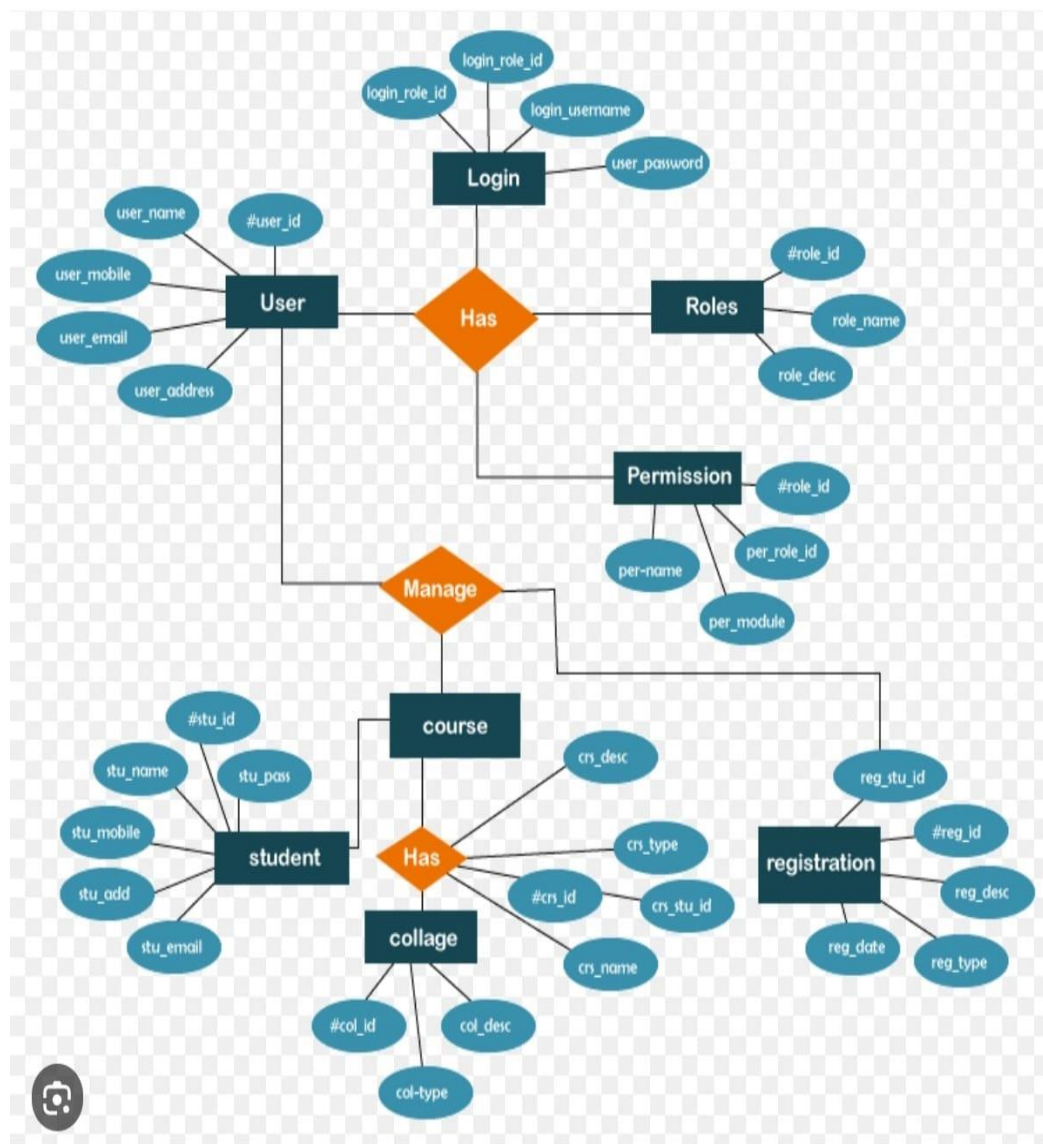


Fig 6.2.4.1: Collabaration Diagram

6.2.5 ACTIVITY DIAGRAM :

An Activity diagram is a type of Unified Modeling Language (UML) diagram that is used to model the dynamic aspects of a system, particularly the flow of activities and actions within a business process, use.

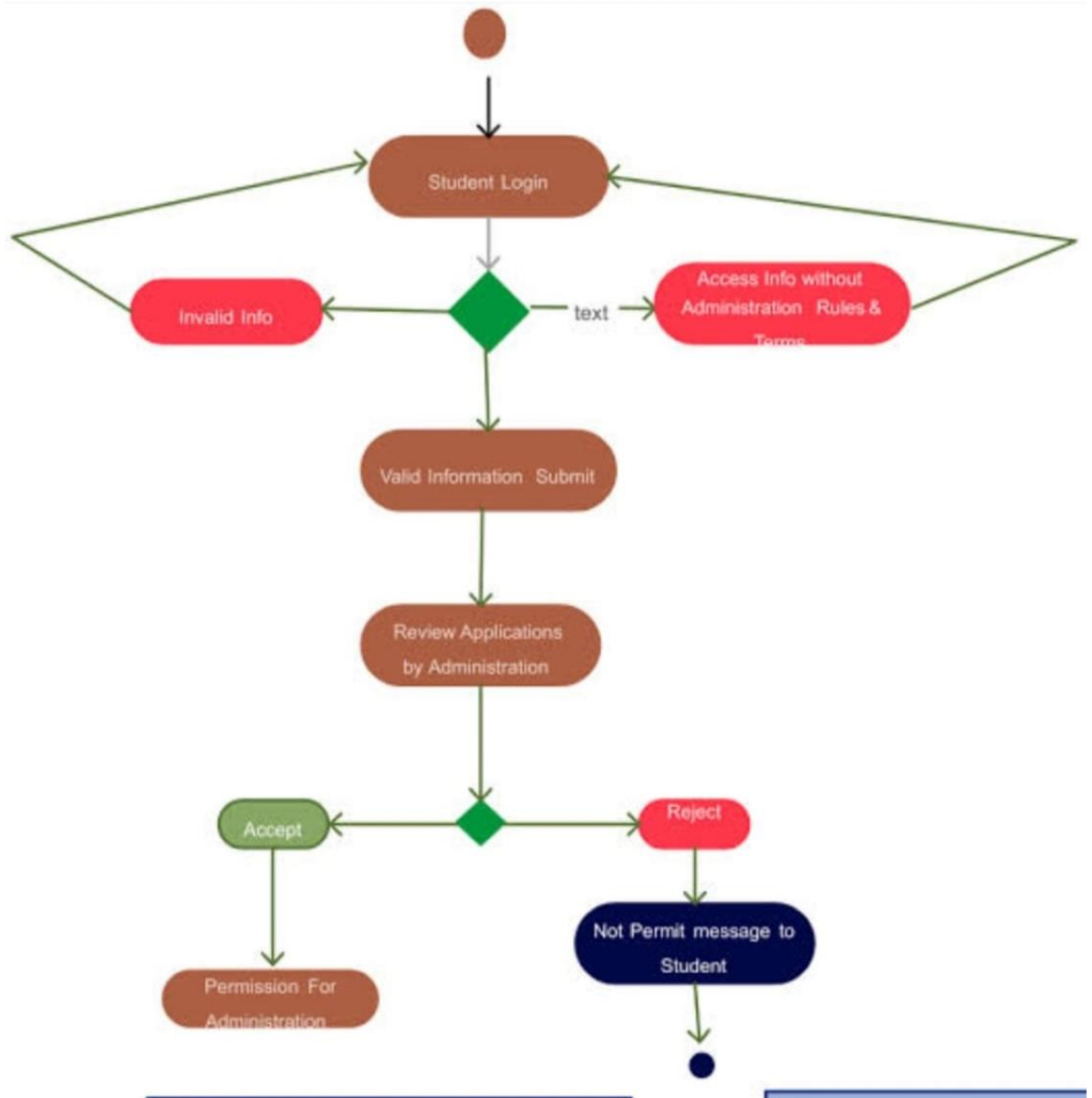


Fig 6.2.5.1: Activity Diagram

CHAPTER-7

INPUT AND OUTPUT DESIGN

7.1 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

OBJECTIVES

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.

3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in a maze of instant. Thus the objective of input design is to create an input layout that is easy to follow

7.2 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displayed for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analyzing design computer output, they should identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- ❖ Convey information about past activities, current status or projections of the
- ❖ Future.
- ❖ Signal important events, opportunities, problems, or warnings.
- ❖ Trigger an action.
- ❖ Confirm an action.

CHAPTER-8

IMPLEMENTATION

8.1 MODULES

- USER
- ADMIN
- COLLEGES
- PAYMENT

8.1.1 MODULE DESCRIPTION

User

This module provides interfaces for users to interact with the system. It can include mobile apps, web portals, or even physical kiosks where users can check availability, reserve parking spots, and make payments.

Admin

This module is used by administrators to monitor system performance, manage parking spaces, update pricing policies, and generate reports on usage statistics.

Colleges

This module is used by user to search their respective colleges.it mainly gives certain different options .which includes different universities from different cities and countries.

Status Module

In this system the status module updates the status of candidate at the time of admission. By default, status of a candidate is made “listed”. At the day of admission, candidate’s status updates as follows:

The status “Listed” is updated as “Reported” by the Reporting Desk if the candidate gives attendance for admission.

The status “Reported” is updated as “Document verified” by Document verifier after verifying the required documents.

The status “Document verified” is updated as “Hostel Allocated” by Hostel Allocator when candidate is allocated Hostel with hostel details.

The status “Hostel Allocated” is updated as “Fees Paid” by Finance Section after the admission fees are paid.

After fees paid, the status is updated as “Admission Confirmed” by Academic Section and generates admission receipt.

Confirmation Module

“Admission Receipt” of the candidate by the Academic Section after admission is confirmed. This receipt contains information such as Form Number, Enrollment Number, Candidate Name, Programme Name, Document list which are not submitted at the time of admission etc.

CHAPTER-9

SOFTWARE ENVIRONMENT

SOURCE CODE:

Home.html

```
{% load static %}

<html>

<head>

    <title>Hassle College</title>

    <style>

        body {

            background-image: url("{% static 'bg.png' %}");

            background-size: cover;

            font-family: Arial, sans-serif;

        }

        #header {

            height: 100px;

            border: 1px solid black;

            display: flex;

            background: rgba(255, 255, 255, 0.5);

        }

        #logo {

            height: 100px;

            width: 10%;

            display: flex;

            align-items: center;

            justify-content: center;

        }

        img {

            clip-path: circle(50%);
```

```

}
#heading {
    height: 100px;
    width: 40%;
    display: flex;
    align-items: center;
    justify-content: center;
}
#navi {
    height: 100px;
    width: 50%;
    display: flex;
    align-items: center;
}
#links ul {
    margin: 0;
    padding: 0;
    list-style: none;
    display: flex;
    width: 100%;
    justify-content: space-around;
}
#links ul li {
    padding: 10px 20px;
    background: rgba(255, 255, 255, 0.5);
    border-radius: 15px;
}
#links ul li:hover {
    background: rgba(255, 255, 255, 0.8);
}

```

```

a {
    text-decoration: none;
    color: black;
}
#body {
    height: 500px;
    display: flex;
    align-items: center;
    justify-content: center;
}
#body1 {
    width: 80%;
    background: rgba(255, 255, 255, 0.5);
    padding: 20px;
    text-align: center;
}
#footer {
    height: 100px;
    background: rgba(255, 255, 255, 0.5);
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 0 20px;
}
#social a img:hover {
    background: rgba(255, 255, 255, 0.5);
    border-radius: 50%;
}
</style>
</head>

```

```

<body>
  <div id="header">
    <div id="logo">
      
    </div>
    <div id="heading">
      <h1>Hassle College</h1>
    </div>
    <div id="navi">
      <div id="links">
        <ul>
          <li><a href="/">Home</a></li>
          <li><a href="HS">Login</a></li>
          <li><a href="HSR">Courses</a></li>
          <li><a href="admissions">Admissions</a></li>
          <li><a href="contact">Contact</a></li>
        </ul>
      </div>
    </div>
  </div>
  <div id="body">
    <div id="body1">
      <p>
        Welcome to Hassle College, where education meets innovation. We provide
        top-notch academic programs to shape the future of our students.
      </p>
    </div>
  </div>
  <div id="footer">
    <div id="social">

```

```
<a href="https://www.facebook.com"></a>

<a href="https://www.instagram.com"></a>

<a href="https://www.linkedin.com"></a>

<a href="https://twitter.com"></a>

<a href="https://youtube.com"></a>

</div>

<div id="address">

    <p>123 College Avenue, Academic City, Earth</p>

</div>

</div>

</body>

</html>
```


CHAPTER-10

RESULTS/DISCUSSIONS

10.1 SYSTEM TESTING:

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS:

UNIT TESTING:-

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

INTEGRATION TESTING:-

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successfully

unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

FUNCTIONAL TEST:-

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

SYSTEM TEST:-

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

WHITE BOX TESTING:-

White Box Testing is a testing in which in which the software tester has

knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

BLACK BOX TESTING:-

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

UNIT TESTING:-

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

TEST STRATEGY AND APPROACH:-

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format

- No duplicate entries should be allowed
- All links should take the user to the correct page.

integration testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing:-

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

10.1.1 TESTCASES:-

Test case1 for Login form:

FUNCTION:	LOGIN
EXPECTED RESULTS:	Should Validate the user and check his existence in database
ACTUAL RESULTS:	Validate the user and checking the user against the database
LOW PRIORITY	No
HIGH PRIORITY	Yes

Test case2:

Test case for User Registration form:

FUNCTION:	USER REGISTRATION
EXPECTED RESULTS:	Should check if all the fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	No
HIGH PRIORITY	Yes

10.2 SCREENSHOTS:

HOME PAGE:

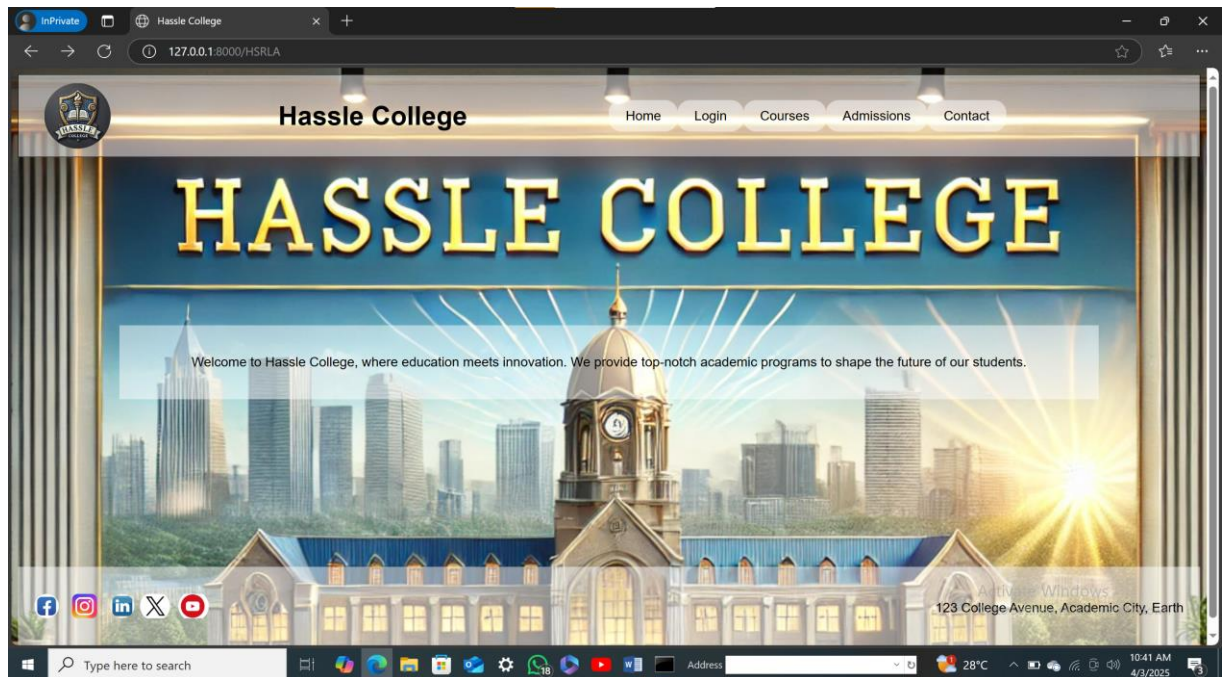


Fig 10.2.1: Home Page

LOGIN PAGE:

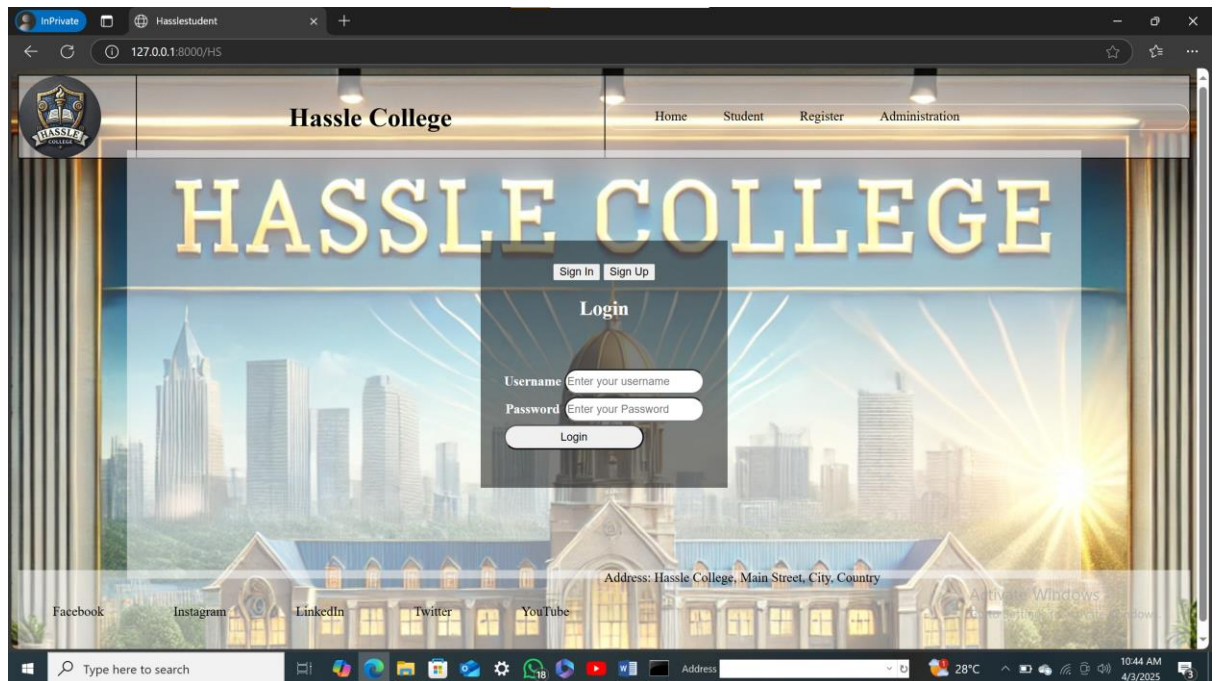


Fig 10.2.2: Login Page

REGISTRATION PAGE:

Hassle College

Home Student Register Librarian

HASSLE COLLEGE

Student Registration

Name

Gender ☐ Male ☐ Female

Language ☐ Hindi ☐ English

Year of Passout

Username

Password

Windows Taskbar: 10:48 AM 4/8/2025, 28°C

Fig 10.2.3: Registration Page

REGISTRATION SUCCESSFUL PAGE:

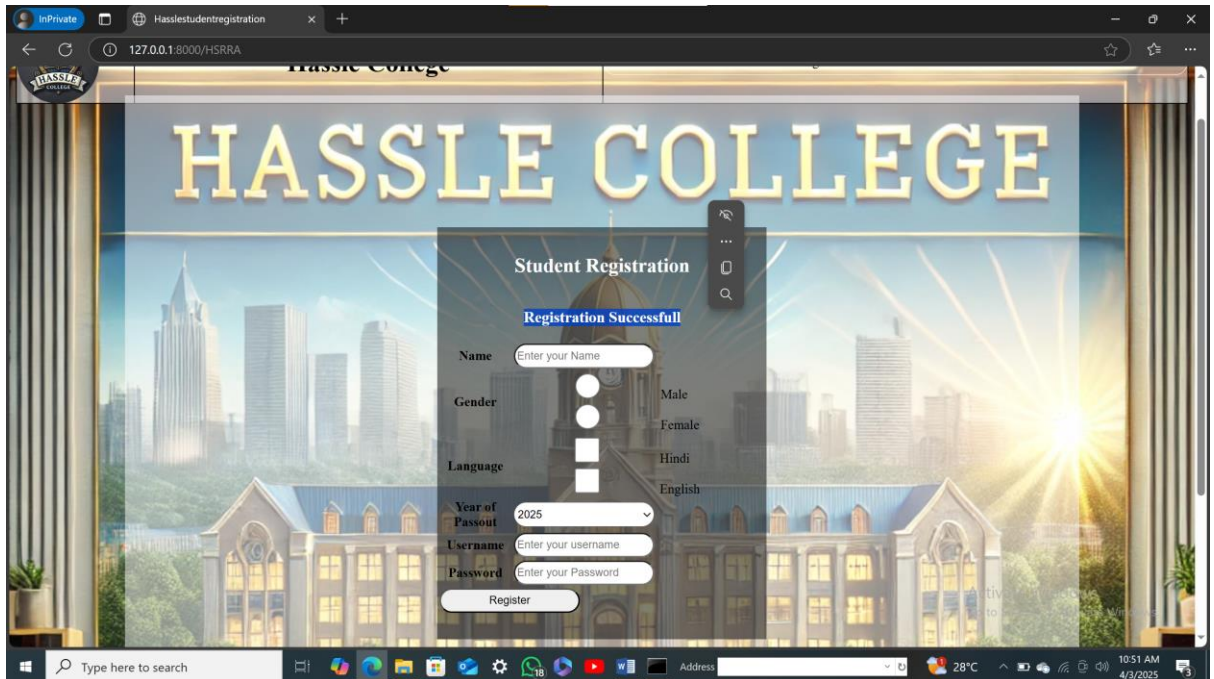


Fig 10.2.4: Registration Successful Page

CHAPTER-11

CONCLUSION

11.1 Conclusion:

The university admission system plays a crucial role in shaping the academic and professional trajectories of students. It serves as a gateway for individuals to access higher education and pursue their chosen fields of study. The process involves a combination of academic performance evaluation, standardized testing, extracurricular achievements, recommendation letters, and personal statements.

Several challenges and debates surround the university admission system, including issues related to equity, access, and fairness. Critics argue that certain admission criteria may perpetuate socio-economic disparities and hinder opportunities for underrepresented groups. As a response, there is a growing emphasis on holistic admissions, which considers a broader range of factors beyond academic metrics.

In recent years, universities and colleges have been exploring innovative approaches to admissions, such as test-optional policies and the use of holistic review processes. These changes aim to create a more inclusive and diverse student body, recognizing that academic achievement alone may not fully capture an individual's potential or contribution to the university community.

Technology has also played a role in the admission process, with the use of online application platforms, virtual interviews, and data analytics to streamline and enhance the efficiency of the system. Additionally, internationalization efforts have led to an increase in the diversity of student populations on campuses worldwide.

In conclusion, the university admission system continues to be a dynamic and sometimes contentious aspect of higher education. Ongoing discussions and reforms

aim to strike a balance between academic rigor and inclusivity, ensuring that deserving individuals have the opportunity to access quality education and contribute to the academic and societal landscape. It is important to stay informed about current trends and changes in the admission process to navigate this critical step in the educational journey.

11.2 Future Scope:

The "Hassle-Free College Admission Portal" has successfully streamlined the admission process for students and administrators. However, there remains considerable potential for further enhancement and scalability. The following future scope outlines possible improvements and additions to the system:

- 1. Integration with national academic databases** for secure document verification.
- 2. AI-powered ChatBot** to assist students with queries.
- 3. Mobile application** for easier access and notifications.
- 4. Automated merit list generation** to streamline selection.
- 5. Multilingual support** to improve accessibility.
- 6. Advanced data analytics** for better administrative decisions.
- 7. Enhanced security** through two-factor authentication.
- 8. Cloud deployment** to ensure scalability during peak times.

CHAPTER-12

REFERENCES

1. Admission procedure & policy (2012). Retrieved January 28, 2012, From <http://www.qou.edu/english/index.jsp?pageId=25>. Alexander, S. et al., (2003), Case Studies in Admissions to and Early Performance in Computer Science Degrees, ITiCSE-WGR '03 Working group reports from ITiCSE on Innovation and technology in computer science education, ACM New York, NY, USA Alsunbul, A. (2002).
2. Issues relating to distance education in the Arab world. *Convergence*. 35(1), 59-81. ERIC Database. Bergsten, H. (2002). *JavaServer Pages 2nd Edition*, Publisher O'Reilly. ISBN 059600317X. Barnum, C. M. (2002). *Usability Testing and Research*, New York Longman. Needham Heights, MA. Allyn & Bacon, Inc. Boyer, J. M. (2008).
3. Interactive Office Documents A New Face for Web 2.0 Applications, Proceedings of the 2008 ACM Symposium on Document Engineering. Caulton, D. A. (2001). Relaxing the homogeneity assumption in usability testing. *Behaviour & Information Technology*, 20 1–7. Cavana, R., Delahaye, B., & Sekaran, U. (2001).
4. *Applied Business Research Qualitative and Quantitative Methods*, John Wiley & Sons, Milton, Queensland. Chen, M. Y., Kıcıman E., & Brewer E. (2003). An Online Evolutionary Approach to Developing Internet Services, HOTOS'03 Proceedings of the 9th conference on Topics in Operating Systems, Vol 9. Clark, M.C & Boyle R. D., (2003).
5. Discipline specific preparation for university study. *Teaching in Higher Education*,. In preparation. Cooper, J. W. (2000). *Java™ Design Patterns A Tutorial*, Addison-Wesley Professional. Fielding, R. T., & Taylor, R. N. (2002). Principled design of the modern Web architecture, *ACM Transactions on Internet Technology (TOIT)*, v.2 n.2, p.115-150. Fan, K., Li N., Wu, Q., & Liu, X. (2010).

6. A framework of on-line office document processing tool based on Xforms, Published in IEEE International Conference on Software Engineering and Service Sciences (ICSESS). 97 Fuggetta, F., Picco, G. P., & Vigna, G. (1998).
7. Understanding Code Mobility, IEEE Transactions on Software Engineering, v.24 n.5, p.342-361. Gaedke, M., & Graef, G. (2000). Development and Evolution of Web- Applications using the Web Composition Process Model. In International Workshop on Web Engineering at the 9th International WWW Conference. Galbraith, J. R. (1979).
8. Organization Design An Information Processing View. In R. C. Huseman (Eds.), Readings in Interpersonal and Organizational Communication Boston, MA Allyn and Bacon. Ghanem, F., & Hamayil, M. (2011). Management Role of Al-Quds Open University in Leading Effective Distance Learning Programs An Evaluation of QOU Experience in Blended Learning, Contemporary educational technology, 2(1), 55-76. Jusuf, S., Nevenka, T., & Miroslav, Z. (2003).
9. University Without Paper”, 25th Int. Conf. IT Interfaces ITI, Cavtat, Croatia. Konishi, K., Furukawa, N., & Ikeda, H. (2007). Data Model and Architecture of a Paper-Digital Document Management System, DocEng’07, August 28-31, Winnipeg, Manitoba, Canada, ACM. Lewis, J. (1995). IBM computer usability satisfaction questionnaires psychometric evaluation